# A COUNTEREXAMPLE TO THE SQUARE CONJECTURE FOR TRIANGULAR MATCHSTICK PRIME SUMS

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ABSTRACT. Let $p_j$ be the $j$th prime, with $p_1 = 2$, and let $a(n) = \sum_{j=n}^{2n} p_j$. The square conjecture associated with OEIS A105720 asserts that, for positive $n$, the value $a(n)$ is a square exactly when $n \in \{3, 6, 4072\}$. We give the counterexample

$$a(111008771) = 387024194082134596 = 622112686^2.$$

The calculation is certified by exact prime counts and prime sums at two explicit bounds. We prove the correctness of the compressed sieve used to obtain these values and provide a single-file Lean refutation of the precise statement in the Formal Conjectures repository. The finite numerical checks use Lean's `native_decide`; its compiler trust assumption is stated explicitly.

## 1. THE CONJECTURE AND THE COUNTEREXAMPLE

The sequence A105720 in the On-Line Encyclopedia of Integer Sequences [1] is defined by

$$a(n) = \sum_{j=n}^{2n} p_j, \qquad n \geq 1. \tag{1}$$

Its entry asks whether the only square terms occur at $n = 3, 6, 4072$. The corresponding values are $6^2$, $13^2$, and $15735^2$. The Formal Conjectures repository [2] expresses the question as the equivalence

$$\forall n \geq 1, \quad a(n) \text{ is a square } \iff n = 3 \text{ or } n = 6 \text{ or } n = 4072. \tag{2}$$

We disprove this equivalence by an additional square term.

**Theorem 1.1.** *For $N = 111008771$,*

$$\sum_{j=N}^{2N} p_j = 387024194082134596 = 622112686^2.$$

*In particular, the equivalence* (2) *is false.*

The result is a finite, computer-assisted disproof. It does not classify all square terms of the sequence or assert that there are infinitely many such terms. The sieve employed below is a compressed form of Eratosthenes' sieve; the contribution here is the explicit counterexample and its verification against the stated conjecture.

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## 2. An exact arithmetic certificate

Write

$$\pi(x) = \#\{q \le x : q \text{ is prime}\}, \qquad \mathscr{P}(x) = \sum_{\substack{q \le x \\ q \text{ prime}}} q.$$

Set $L = 2274652151$ and $U = 4711447171$. The certificate consists of the following integers:

| $x$ | $\pi(x)$ | $\mathscr{P}(x)$ |
|---|---|---|
| $2274652150 = L-1$ | 111008770 | 123000804558174777 |
| $4711447171 = U$ | 222017542 | 510024998640309373 |

The endpoints $L$ and $U$ are also prime. Primality can be checked by trial division through their integer square roots. Sections 3 and 4 justify and certify the four table entries.

*Proof of Theorem 1.1 from the certificate.* The identity $\pi(L-1) = N-1$, together with the primality of $L$, gives $L = p_N$. Similarly, $\pi(U) = 2N$ and the primality of $U$ give $U = p_{2N}$. Therefore

$$\begin{aligned} a(N) &= \mathscr{P}(U) - \mathscr{P}(L-1) \\ &= 510024998640309373 - 123000804558174777 \\ &= 387024194082134596 \\ &= 622112686^2. \end{aligned}$$

Since $N$ is positive and differs from 3, 6, and 4072, this square term contradicts the forward implication in (2). □

## 3. Correctness of the compressed sieve

For an integer $m \ge 2$, let $\ell(m)$ denote its least prime factor. Before processing an integer $p \ge 2$, define the survivors below $v$ by

$$R_p(v) = \{m : 2 \le m \le v, \quad m \text{ is prime or } p \le \ell(m)\}.$$

Let $C_p(v) = \#R_p(v)$ and $T_p(v) = \sum_{m \in R_p(v)} m$. For $v \ge 1$, the initial state is

$$C_2(v) = v - 1, \qquad T_2(v) = \frac{v(v+1)}{2} - 1. \tag{3}$$

**Lemma 3.1.** *If $p$ is prime and $v \ge p^2$, then*

$$C_{p+1}(v) = C_p(v) - \big(C_p(\lfloor v/p \rfloor) - C_p(p-1)\big), \tag{4}$$

$$T_{p+1}(v) = T_p(v) - p\big(T_p(\lfloor v/p \rfloor) - T_p(p-1)\big). \tag{5}$$

*If $p$ is composite, or if $v < p^2$, then $R_{p+1}(v) = R_p(v)$.*

*Proof.* An integer belongs to $R_p(v) \setminus R_{p+1}(v)$ precisely when it is composite and its least prime factor is $p$. Such an integer has the form $pm$, where

$$p \le m \le \lfloor v/p \rfloor, \qquad p \le \ell(m).$$

Conversely, these conditions imply that $pm$ is composite with least prime factor $p$. Indeed, a prime divisor of $pm$ either equals $p$ or divides $m$, and in the latter case it is at least $p$. Thus multiplication by $p$ is a bijection

$$R_p(\lfloor v/p \rfloor) \setminus R_p(p-1) \longrightarrow R_p(v) \setminus R_{p+1}(v).$$

Here a surviving prime $m \ge p$ also satisfies $\ell(m) = m \ge p$. The hypothesis $v \ge p^2$ ensures $R_p(p-1) \subseteq R_p(\lfloor v/p \rfloor)$. Taking cardinalities and sums gives (4) and (5).

If $p$ is composite it cannot be a least prime factor. If $v < p^2$, there is no composite below $v$ with least prime factor $p$, since every composite $r$ satisfies $\ell(r)^2 \le r$. These observations give the two unchanged-state cases. □

For a fixed bound $x$, put $B = \lfloor \sqrt{x} \rfloor$ and use the finite set

$$V_x = \{\lfloor x/i \rfloor : 1 \le i \le B\} \cup \{1, \dots, B\}. \tag{6}$$

This uses at most $2B$ entries. The entries needed on the right of (4) and (5) are present in this set.

**Lemma 3.2.** *If $v \in V_x$, $2 \le p \le B$, and $v \ge p^2$, then $\lfloor v/p \rfloor \in V_x$ and $p-1 \in V_x$.*

*Proof.* The second assertion is immediate. If $v \le B$, then $1 \le \lfloor v/p \rfloor \le B$. Otherwise write $v = \lfloor x/i \rfloor$ with $i \le B$. Integer division gives

$$\lfloor v/p \rfloor = \lfloor x/(ip) \rfloor .$$

If $ip \le B$, this is an entry in the quotient part of $V_x$. If $ip > B$, then $x < (B+1)^2$ gives $\lfloor x/(ip) \rfloor \le B$, so it is in the small-integer part. The assumption $v \ge p^2$ ensures the quotient is positive. □

**Corollary 3.3.** *Initialize the tables on $V_x$ by* (3)*, and apply Lemma 3.1 for $p = 2, 3, \dots, B$. Read each right-hand side from the preceding stage. The final values at $x$ are exactly $\pi(x)$ and $\mathscr{P}(x)$.*

*Proof.* Lemmas 3.1 and 3.2 maintain the invariant that the tables equal $C_p$ and $T_p$ at every stored bound. After processing $B$, the survivors are $R_{B+1}(v)$. If a composite $m \le x$ survived, it would satisfy $\ell(m) \ge B+1$, hence $m \ge (B+1)^2 > x$, a contradiction. Every prime survives every stage. The final survivors below $x$ are therefore precisely the primes at most $x$. □

The two certificate computations use $B = 47693$ and $B = 68639$, respectively. All calculations use exact integer arithmetic. The independent Python verifier reads and updates the compressed tables in decreasing order of $v$. Since $\lfloor v/p \rfloor < v$, the required entries still hold the preceding stage's values at the time of each update. The Lean implementation instead reads an immutable preceding-stage table and writes the new entries to a separate logical table; both implement the same recurrences.

## 4. Formal refutation and reproducibility

The companion file `TriangularMatchstickPrimeSums.lean` imports the exact repository module

`FormalConjectures.OEIS.«105720»`.

The repository uses zero-based indexing for `Nat.nth Nat.Prime`. Its definition, for $n > 0$, is

$$\texttt{OeisA105720.a}(n) = \sum_{i \in [n-1,\, 2n)} \texttt{Nat.nth Nat.Prime}(i),$$

which is exactly (1). In particular, there are $n+1$ summands, including both endpoint primes.

The formal development proves the survivor characterization, the multiplication bijection of Lemma 3.1, the weighted recurrence, and the table invariant. Array indexing and the finite grid's closure conditions are checked as explicit decidable propositions. The computed tables are then connected to Mathlib's finite sets of primes. Finally, the bijection between those sets and the initial segments of `Nat.nth Nat.Prime` identifies the prime-prefix sums with the repository's indexed sum.

The concluding statements have the following form:

```
theorem counterexample_value :
  OeisA105720.a 111008771 = 622112686 ^ 2

theorem conjecture_false :
  Not (forall (n : Nat), 0 < n ->
    (IsSquare (OeisA105720.a n) <->
      Or (n = 3) (Or (n = 6) (n = 4072))))
```

The displayed second statement uses Lean's ASCII notation; the companion file uses the corresponding mathematical symbols. The original repository declaration of the conjecture is never used as a premise.

**The trust boundary.** The large finite table calculations and closure checks use `native_decide`. This tactic trusts Lean's evaluation of compiled code [3]; it is not a certificate whose entire numerical computation is reduced by the logical kernel. In Lean 4.33.1 these checks introduce automatically generated evaluation axioms. The final theorems depend on the four such axioms for `lower_grid`, `lower_result`, `upper_grid`, and `upper_result`, together with the standard axioms `propext`, `Classical.choice`, and `Quot.sound`. The source includes `#print axioms` checks for the final theorems so that this dependency is visible. There are no admitted proofs in the companion file, and the proof does not depend on the unproved repository conjecture. The independent exact Python and C++ calculations provide additional checks of the numerical inputs, while the generic sieve lemmas and the connection to the repository statement are proved in Lean.

**Pinned environment.** The development targets Lean 4.33.1 and the Formal Conjectures revision

`89294ea02bd7cd678d59984add52cb4baef3dbf4.`

That revision pins Mathlib to

`0df444a360eaa60ab8c11dca51a86af692955474.`

After building `FormalConjectures.OEIS.105720` in that checkout, the single companion file can be checked from the repository root with

```
lake env lean /absolute/path/to/TriangularMatchstickPrimeSums.lean
```

The accompanying `verify_a105720.py` independently verifies the certificate and the square identity. An exact segmented-sieve search in `search_a105720.cpp` located the counterexample using

$$a(n+1) = a(n) - p_n + p_{2n+1} + p_{2n+2}, \qquad a(1) = 5.$$

The search is a discovery tool; the disproof needs only the two prime-count and prime-sum evaluations in Section 2.

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## REFERENCES

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[3] The Lean Language Reference, *Tactic reference*, `native_decide`, version 4.33. `https://lean-lang.org/doc/reference/4.33.0/Tactic-Proofs/Tactic-Reference/`.

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